ITO and Pt Deposition for Improved Performance of Neural Probes.

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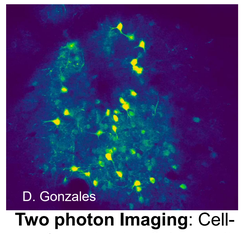
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Introduction & Background

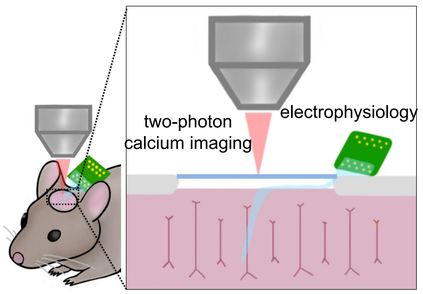


Neural electrophysiology: Millisecond-scale activity with high temporal resolution

Integrating electrophysiology with optogenetics enables **simultaneous recording and stimulation of neural circuits**, but traditional metal electrodes are **opaque and incompatible with efficient light delivery**

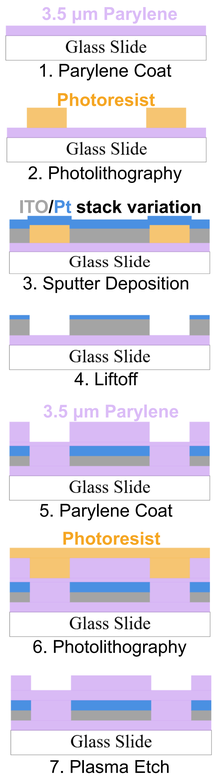


Two photon Imaging: Cell-specific and subcellular spatial and structural dynamics

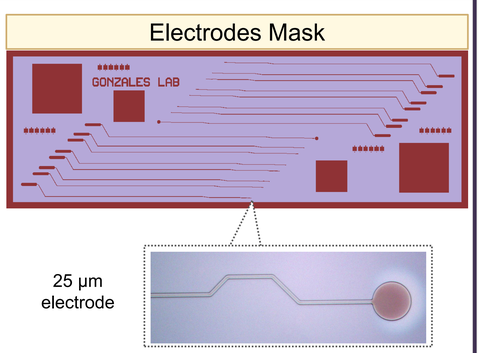


Solution: Implantable probes made with a conductive stack of **ITO and Pt** that allow for simultaneous in vivo electrophysiology and two-photon imaging

Fabrication Process

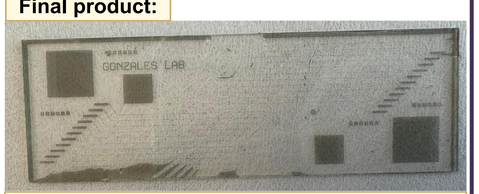


Electrodes Mask



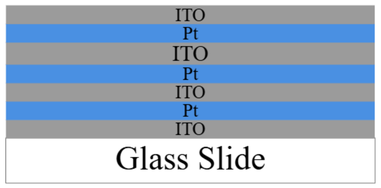
Photolithography mask with electrode diameters ranging from 25-500 µm

Final product:

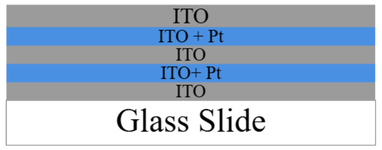


Glass slide with electrodes pattern

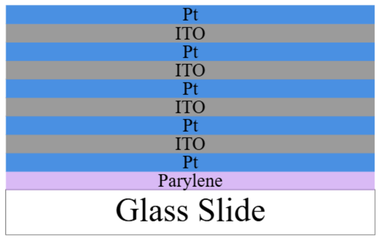
Effect of Deposition Conditions on Film Properties



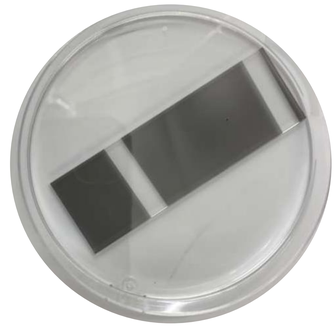
Stack 1
ITO: 70 nm
Pt: 1.95 nm



Co-Deposition 1
ITO: 95.2 nm
Pt: 5.2 nm

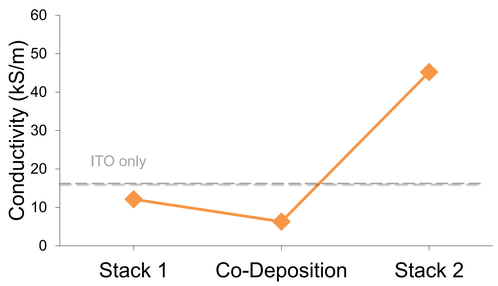


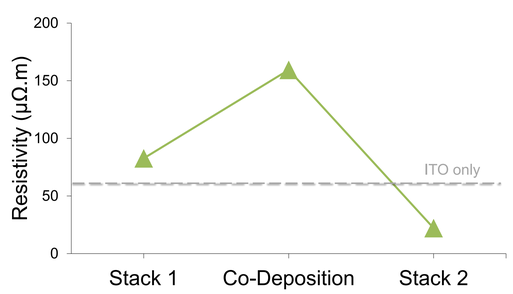
Stack 2
Parylene: 3.5 µm
ITO: 56 nm
Pt: 5.2 nm

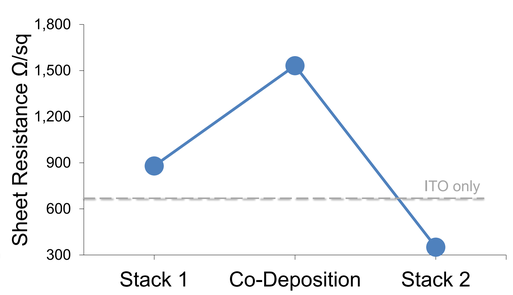


Final Unpattern Stack:

Four-Point Probe Testing



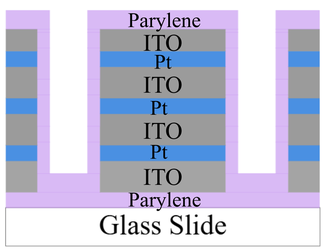




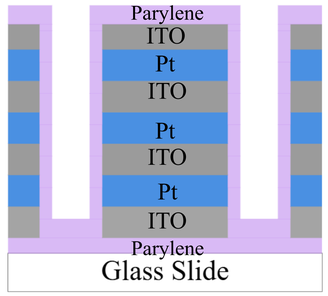
This suggests that layer interfaces and ordering play a critical role in charge transport.

Stacks starting and ending with Pt show improved conductivity.

Patterned Stack

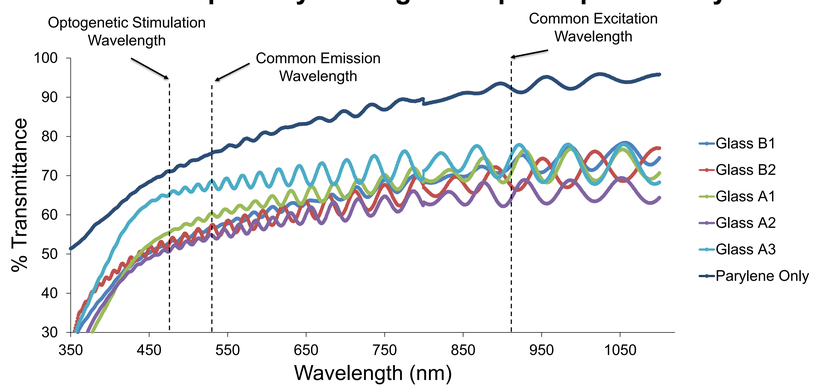


Stack A
ITO: 84 nm
Pt: 1.95 nm



Stack B
ITO: 84 nm
Pt: 5.1 nm

Transparency Testing with Spectrophotometry



Conclusion & Future Directions

Co-deposition resulted in the lowest conductivity, while multilayer stacks—especially those initiated and terminated with Pt—showed the highest performance. This demonstrates a clear dependence of electrical performance on **layer ordering and structure**, with increased Pt thickness further enhancing conductivity. Among the tested configurations, optimized stack 2 achieved **lower resistivity and sheet resistance compared to ITO alone.** These results highlight multilayer stacks as **promising materials for neural probes.**

***Future Work* includes:**

- Perform in-depth mechanical characterization (flexibility, adhesion, and durability) to evaluate suitability for implantation
- Explore alternative deposition parameters or materials to further improve performance

Acknowledgments

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